

GolfIt Game Manual

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1. Game Vision

GolfIt is a real-time multiplayer game that emphasizes physics-based precision and environmental strategy. Unlike traditional turn-based golf, all players in a lobby compete simultaneously in a live environment. The server maintains a high-frequency physics simulation (40ms ticks) to ensure synchronized movement for all participants.

2. Getting Started

2.1 Have a computer

To be able to play *GolfIt*, you first need a working computer, you can buy it if physical stores or online. Important requirement are Java 25 and a display resolution of at least 1280 x 720 pixels. Very important, to be able to play the game, **the computer has to be turned on.**

2.2 Launching the Application

GolfIt is packaged as a single executable JAR file. You must start a server instance before clients can connect. Use the following commands in your terminal:

To start the Server: Run the application with the **server** argument followed by your desired port number.

```
java -jar golfit-0.0.1-ALPHA.jar server <port>
```

To start a Client: Run the application with the **client** argument, providing the server's **address:port** and your chosen optional **username**.

```
java -jar golfit-0.0.1-ALPHA.jar client <ipAdress>:<port> <name>
```

2.3 Joining and Starting a Match

Once the client has successfully connected to the server, the interface displays the Lobby Menu. Players interact with the UI to enter and begin a match:

- **Finding a Match:** The "Lobbies" list displays all available game sessions, including the host's name and the current player count. To enter a game, simply click the **Join** button next to an open lobby.
- **Creating a Lobby:** If no lobbies are available, click the **Create Lobby** button to become the host of a new session.

- **The Waiting Room:** Once inside a lobby, you can see the names of other participants and the player suggested limit. In this view, you can use the integrated chat to coordinate with others. Your unique ball color is automatically determined by your position in the join order.
- **Start the Game:** To start the game, every lobby member can press the **START** button. This action locks the lobby and transitions all connected players to the the game.
- **Spawning:** Upon starting, all players spawn simultaneously at the starting tee. The server automatically calculates staggered coordinates for each ball to ensure that no players overlap at the beginning of the race.

3. Controls and Physics

Players interact with their ball using a "Stroke" mechanic that applies impulse forces.

- **Aiming:** Players set a direction and power (visualized via a multi sections colored cone).
- **Bouncing ball:** The golf ball bounce on physical obstacles in the map (other golf balls not considered). The bouncing logic respect the real physics: the angle of incidence is the same of the angle of reflection.
- **Stopping Velocity:** A ball is considered "stopped" and ready for the next move only when its velocity drops below 1.5 grid/s.

4. Terrain and Environmental Hazards

The course contains various tile types that directly impact ball physics through friction and collision.

- **Grass (Standard):** Balanced friction for predictable movement.
- **Sand:** High resistance with high friction, making it difficult to maintain momentum.
- **Ice:** Extremely low resistance with low friction, causing the ball to slide significantly further.
- **Water:** Contact with water or falling off the map resets the ball to its last stable position.

5. Interactive Mechanics

Maps feature dynamic obstacles managed by the server's physics engine:

- **Switch Triggers:** Rolling over a switch triggers a global environmental change.
- **Temporal Toggles:** Activating a switch despawns the bridge and spawns a barrier for 5 seconds before reverting to their original state.

6. Win/Loss Conditions

- **Hole Capture:** The goal is to reach the black hole (in the right up corner).
- **Win Condition:** The winner is the player that captures the hole with the least stroke, in case of tie, player who finished first wins.